

Device Lock Portal API

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History

Status/Version/Date	Author	Comment
Stable/1.0.0/2026-07-15	Lei.Gao	First version.
Stable/1.0.1/2026-07-29	Lei.Gao	Corrected API specifications and improved descriptions.

1 Scope

This document describes the Portal APIs of the WhaleOS Device Lock Management Service.

2 Architecture

These Open APIs provide functionality consistent with the Device Lock Portal.

3 Specification

The Device Lock Portal API follows standard SaaS API specifications, including URL format, authentication mechanism, and response format.

3.1 URL

API URLs follow the following specifications:

```
https://<host>/lock-portal/open/<api-version>/<api-path>
```

Hosts are listed below:

Environment	Host
ACC	acc-lockservice.whaletv.com
PROD	lockservice.whaletv.com

API Version is currently fixed as: v1

Example:

```
https://acc-lockservice.whaletv.com/lock-portal/open/v1/devices/import
```

3.2 Authentication

Each integration partner needs to apply for a pair of credentials in the company's permission management system:

field	Comment
accessKey	Public identifier used for authentication
secretKey	Secret key that must be stored securely and is used to calculate the signature.
brandId	Brand ID

Include the following request header:

Authorization: <accessKey>:<signature>:<timestamp>

Where:

accessKey: AccessKey obtained through application

timestamp: The current timestamp (in milliseconds) should not deviate from the server time by more than 1 hour

signature: Calculated using the following algorithm

Signature algorithm:

signature = Base64(HmacSHA1(secretKey, requestURI + timestamp))

requestURI: Request path, without domain name or query string, for example:

`/lock-portal/open/v1/devices/import`

The server recalculates the signature according to the same rules and compares it. If there is any inconsistency, it will directly return 401

Signature Example(Java):

```
String accessKey = "your-access-key";
String secretKey = "your-secret-key";
String requestURI = "/lock-portal/open/v1/devices/import";
long timestamp = System.currentTimeMillis();

String data = requestURI + timestamp;
Mac mac = Mac.getInstance("HmacSHA1");
mac.init(new SecretKeySpec(secretKey.getBytes(StandardCharsets.UTF_8), "HmacSHA1"));
byte[] signBytes = mac.doFinal(data.getBytes(StandardCharsets.UTF_8));
String signature = Base64.getEncoder().encodeToString(signBytes);

String authorization = accessKey + ":" + signature + ":" + timestamp;
// Add the following request header: Authorization: <authorization>
```

Authentication Failure Response:

HTTP status: 401

Content-Type: text/html

Message	Comment
AUTHORIZATION EMPTY	Authorization header is missing
AUTHORIZATION FORMAT ERROR	Format error
AUTHORIZATION TS ERROR	Timestamp deviation exceeding 1 hour
AUTHORIZATION ACCESSKEY NON-EXISTENT	AccessKey does not exist
AUTHORIZATION SIGNATURE ERROR	Signature verification failed
USERINFO IS NULL	No API permission

3.3Response

Successful response:

```
{
  "errorCode": 0,
  "errorMsg": "success",
  "timestamp": 1780000000000,
  "data": {}
}
```

Error response:

```
{
  "errorCode": 270201,
  "errorMsg": "parameter invalid",
  "timestamp": 1780000000000
}
```

Numeric and Boolean values may be serialized as JSON strings.

4 API List

4.1Get Device List

API Path	/open/v1/devices
Method	GET
Description	Retrieve a paginated list of devices

Parameters: Query String

Name	Required	Comment
brandId	Y	Brand ID
mac	N	MAC address
sn	N	serial number
status	N	Lock status: 0-Unlocked, 1-Locked
nextInstallmentDate	N	Next installment payment date, yyyy-MM-dd
dueStatus	N	Due status: 0-Not overdue, 1-Overdue
paymentStatus	N	Payment status: 0-In Progress, 1-Complete
pageNum	N	Page number, default value: 1
pageSize	N	Number of items per page, default value: 20, pageSize maximum:

		200
--	--	-----

Response:

Field	Type	Description
id	Long	Device ID
mac	String	MAC address
sn	String	serial number
status	Integer	Lock status: 0-Unlocked, 1-Locked
lastOnlineTime	Long	Last online time of the device, millisecond timestamp
importTime	Long	Import time, millisecond timestamp
nextInstallmentDate	String	Next installment payment date, yyyy-MM-dd
dueStatus	Integer	Due status: 0-Not overdue, 1-Overdue
paymentStatus	Integer	Payment status: 0-In Progress, 1-Complete

Page Response example:

<pre>{ "errorCode": 0, "errorMsg": "success", "timestamp": 1780000000000, "totalSize": 100, "data": [] }</pre>
--

4.2 Device Detail

API Path	/open/v1/devices/detail
Method	GET
Description	Get device details

Parameters: Query String

Name	Required	Comment
id	Y	Device ID
brandId	Y	Brand ID

Response:

Field	Type	Description
id	Long	Device ID
brandId	Long	Brand ID
mac	String	MAC address
sn	String	serial number
status	Integer	Lock status: 0-Unlocked, 1-Locked
lastOnlineTime	Long	Last online time of the device, millisecond timestamp

importTime	Long	Import time, millisecond timestamp
nextInstallmentDate	String	Next installment payment date, yyyy-MM-dd
dueStatus	Integer	Due status: 0-Not overdue, 1-Overdue
paymentStatus	Integer	Payment status: 0-In Progress, 1-Complete
installmentCycle	String	Installment cycle in months
curPincode	String	The currently generated PIN code
curPasscode	String	The currently generated passcode
clearStatus	Integer	Clear status: 0-Default, 1-Clearing

4.3 PIN code history

API Path	/open/v1/devices/pincode-history
Method	GET
Description	Get PIN code usage history

Parameters: Query String

Name	Required	Comment
id	Y	Device ID
brandId	Y	Brand ID

Response:

Field	Type	Description
passcode	String	passcode
pincode	String	PIN code
date	Long	Usage date, millisecond timestamp
status	Integer	PIN code status: 0-Not used, 1-Issued, 2-Used

4.4 Batch Import Devices

API Path	/open/v1/devices/import
Method	POST
Content-Type	application/json
Description	Batch import devices

Parameters:

Name	Required	Comment
brandId	Y	Brand ID
isSn	N	Device identifier type: 0-MAC, 1-serial number. Default: 0
items	Y	String[]

Device identifier list, with a maximum of 1000 entries per request

Example:

```
{
  "brandId": 100,
  "isSn": 0,
  "items": ["AA:BB:CC:DD:EE:01", "AA:BB:CC:DD:EE:02"]
}
```

Response:

Field	Type	Description
success	Integer	Number of successful imports
fail	Integer	Number of failed imports
batchId	Long	Import Batch ID

4.5 Update Device

API Path	/open/v1/devices/update
Method	POST
Content-Type	application/json
Description	Modify the information of a device

Parameters:

Name	Required	Comment
brandId	Y	Brand ID
id	Y	Device ID
status	N	Lock status: 0-Unlocked, 1-Locked
paymentStatus	N	Payment status: 0-In Progress, 1-Complete
nextInstallmentDate	N	The next installment payment date, yyyy-MM-dd
installmentCycle	N	Installment cycle (months)

At least one of status, paymentStatus, nextInstallmentDate, and installmentCycle must be provided.

Example:

```
{
  "brandId": 100,
  "id": 1234567890,
  "status": 1,
  "paymentStatus": 0,
  "nextInstallmentDate": "2025-12-10",
  "installmentCycle": 3
}
```

Response:

data: true/false

4.6 Batch Lock/Unlock

API Path	/open/v1/devices/batch-lock
Method	POST
Content-Type	application/json
Description	Batch Lock/Unlock

Parameters:

Name	Required	Comment
brandId	Y	Brand ID
macs	Y	String[] MAC address list, with a maximum of 1000 entries per request
status	Y	Lock status: 0-Unlocked, 1-Locked

Example:

```
{
  "brandId": 100,
  "macs": ["AA:BB:CC:DD:EE:01", "AA:BB:CC:DD:EE:02"],
  "status": 1
}
```

Response:

data: true/false

4.7 Generate PIN Code

API Path	/open/v1/devices/pincode
Method	POST
Content-Type	application/json
Description	Generate PIN Code

Parameters:

Name	Required	Comment
brandId	Y	Brand ID
mac	Y	Device MAC address
passcode	Y	Passcode generated by the device

Example:

```
{
  "brandId": 100,
  "mac": "AA:BB:CC:DD:EE:FF",
  "passcode": "ABCD1234"
}
```

```
}

```

Response:

Field	Type	Description
pincode	String	Generated PIN code

4.8Clear Device

API Path	/open/v1/devices/clear
Method	POST
Content-Type	application/json
Description	Clear device information

Parameters:

Name	Required	Comment
brandId	Y	Brand ID
id	Y	Device ID

Example:

```
{
  "brandId": 100,
  "id": 1234567890
}
```

Response:

data: true/false

5 Error Code List

Error code and description information

ErrorCode	Description
270201	Required parameters are missing or illegal
270202	The brandId is not authorized for this accessKey
270203	Batch operation quantity exceeds the limit
270102	Device does not exist
270104	MAC address format error
270103	No available PIN code